Discrete Biquaternionic Double Spacetime: Rigorous Proofs of Diophantine Conjectures via Torsional Mismatch and PGA Embedding

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Abstract

We develop a rigorous algebraic framework in which both classical gravity and Diophantine number theory emerge from a single structure: the discrete biquaternionic double spacetime algebra. By embedding the 16-dimensional biquaternion algebra $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong \mathrm{Cl}(3,1)$ into the projective geometric algebra $G(3,1,1)$ and discretizing the dual-spacetime rapidities to a finite torus $(\mathbb{Z}/N\mathbb{Z})^6$, we obtain a theory with a strictly bounded normalized torsional scalar $|J_{\mathrm{norm}}| \leq 1$ on admissible configurations (equivalently $|J| \leq 3\pi^2/8$). The bound follows from an elementary estimate on the Killing quadratic form together with non-negative rapidities and anti-synchronisation.

Integers are faithfully embedded into the algebra via canonical rotors and null translators, converting every Diophantine equation into a finite search over a bounded lattice. Chapter 5 develops the dual-axis geometric seat of a Fermat triple: mixed seeds leak the additive null axis into the time-null generator, and on a putative positive solution of degree $n \geq 3$ that leak persists at finite time as a harmonic oscillation of N_1 whose remainder is a nonzero multiple of N_0. The same seat writes a positive power sum as the scalar $\exp(n\alpha)(\cos^n \beta + \sin^n \beta)$. Equal real legs realise that scalar with a positive boost, so the equation and the remainder coexist over the reals. Degree $n \geq 3$ excludes a positive integer solution in which one base divides the other. On a positive integer solution the boost is half the logarithm of one plus the positive integer excess $a^2 + b^2 - c^2$ over c^2. An odd prime exponent dividing none of three coprime positive bases makes $a+b$, $c-a$, and $c-b$ pairwise coprime pure powers summing to twice a composite third base. Incompatibility of the mixed motor with a positive integer power sum remains open, and classical Fermat's Last Theorem is not claimed. On the equal-odd Gaussian leaf of odd exponent $e \geq 5$, the square-difference legs are an admissible pure dual rotor, while the generator's boost $\frac{e-2}{2} \log c$ exceeds a right angle and places that seat outside the admissible cone. Chapters 6–10 outline a common amplification programme for Beal's conjecture, the abc conjecture, the Riemann Hypothesis, Goldbach's conjecture, Polignac's conjecture, and the Collatz conjecture. The problem-specific bridges that would turn those programmes into unconditional theorems are not established, and no unconditional statement of those conjectures is claimed. On the Collatz map the admissible cone is the stable phase of the integer boost and contains only 1 and 2. The shell between that cone and unit normalised height contains only 3 and is narrower than one halving, so the return $4 \to 2$ enters the cone from the region where the height exceeds one. Every admissible abc triple has quality at most that of $1 + 8 = 9$. Every starting value up to 2^{20} reaches 1, and any other cycle would have normalised height greater than 100.

The framework recovers the Teleparallel Equivalent of General Relativity exactly in the continuum limit $N \to \infty$, while the discrete structure supplies both a number-theoretic foundation and a natural ultraviolet cutoff that eliminates singularities and divergences. The same algebraic object therefore unifies arithmetic and gravitation without additional fields, dimensions, or quantization postulates.

The theory yields concrete physical implications, including discrete gravitational-wave echoes, a finite number of admissible nuclear rotor classes at each compactification N, and discrete torsional corrections to nuclear energy levels. A numerical cutoff $A_* \lesssim 300$ is an underived heuristic, not a theorem.

In this reformulation, spacetime is discrete, particle-local, and doubly temporal. Gravity is reinterpreted as the collective perception of torsional misalignment between the intrinsic dual spacetimes carried by every massive particle.

1 Introduction

The Dual Spacetime Theory encodes gravity and inertia as torsional mismatch between two intrinsically paired Minkowski structures carried by every massive particle, realized within the 16-real-dimensional

biquaternion algebra $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong \mathrm{Cl}(3,1)$. The parameter-space torsional mismatch scalar

$$J = \frac{1}{2} \sum_{a=1}^{3} (\alpha_a^2 - \beta_a^2),$$

built from the six-generator Lorentz sector $\mathfrak{so}(3,1)$ (not the twelve-dimensional sum $\mathfrak{so}(3,1) \oplus \mathfrak{so}(3,1)$), is identified with the teleparallel scalar only after a rotor-to-tetrad dictionary is fixed. Variational equivalence with the Einstein–Hilbert action is taken from the TEGR literature and is not re-derived here. Under the half-angle expansion one has $B(\Omega_{\mathrm{biv}}, \Omega_{\mathrm{biv}}) = 2 \sum_a (\alpha_a^2 - \beta_a^2)$, so J is four times $\frac{1}{16} B(\Omega_{\mathrm{biv}}, \Omega_{\mathrm{biv}})$.

To elevate the framework to a rigorous number-theoretic setting, we introduce two decisive extensions. First, the natural compactness induced by the dual map $X \mapsto Xi$ is formalized by discretizing the rapidity parameters:

$$\omega_a = \frac{2\pi n_a}{N}, \qquad \phi_a = \frac{2\pi m_a}{N}, \quad n_a, m_a \in \mathbb{Z},$$

where $N \gg 1$ is the effective compactification parameter. The resulting phase space is the finite torus $(\mathbb{Z}/N\mathbb{Z})^6$. What is finite is the discrete rotor image of this torus, not the unit group of an unspecified integer biquaternion order. Diophantine problems become finite searches over that image, once restricted to the admissible cone.

Second, we embed the biquaternion algebra into the projective geometric algebra $G(3,1,1)$ by adjoining a null vector e_4 ($e_4^2 = 0$). This produces a ten-dimensional bivector space consisting of three hyperbolic generators, three cyclic generators, and four null generators $N_\mu = e_4 \wedge e_\mu$. The null sector supplies a canonical algebraic realization of translational (additive) degrees of freedom, while the extended invariant $J^{(5)}$ remains non-degenerate. The plane-based bracket products of $G(3,1,1)$ further identify the dual spacetime with the normal bundle to three-blades, furnishing a geometric bridge between additive and multiplicative structures.

The dagger operation $R^{\dagger}_{\mathrm{usual}}$ on the usual-sector rotor extracts the dual-sector data from the combined mismatch (equivalently, swaps the rapidity pairs $(\alpha_a, \beta_a) \mapsto (\beta_a, \alpha_a)$), while subsequent reconstruction by the dual rotor recovers the torsional configuration up to sign of J. Decidability of the discrete Diophantine problems then follows from the finiteness of the admissible torus together with either exhaustive lattice search or amplification contradictions against $|J_{\mathrm{norm}}| \leq 1$ (Chapter 10); dagger itself does not decrease the torsional height.

Within this unified algebraic framework the seven classical problems occupy two distinct seats. Fermat's Last Theorem is attached to a dual-axis torsion seed of the projective triple $[a : b : c]$ (Chapter 5): degree $n = 2$ is pure cyclic, degree $n \geq 3$ is mixed, and mixed seeds carry off-axis interference invisible to the scalar J. Already at first order they leak the additive null axis N_1 into the time-null generator N_0. On a putative positive solution the characteristic is elliptic, so the leak survives at finite time as a harmonic sandwich of N_1 whose remainder is a nonzero multiple of N_0 precisely when $\alpha > 0$. Over the reals that remainder coexists with the power sum. The obstruction is integral: degree $n \geq 3$ excludes a positive solution in which one base divides the other, and an odd prime dividing none of three coprime bases forces $a+b$, $c-a$, and $c-b$ to be pairwise coprime pure powers summing to twice a composite third base. The mixed-motor residual remains open, and classical Fermat's Last Theorem is not claimed. Beal's conjecture, the abc conjecture, the Riemann Hypothesis, Goldbach's conjecture, Polignac's conjecture, and the Collatz conjecture remain amplification programmes on the finite torus, reducing respectively to unequal-exponent mismatch, quality bounds, ensembles, decompositions, even-gap resets, or a pure-boost orbit whose stable phase, the admissible cone, contains only 1 and 2. Those bridges remain unproved, and unconditional statements of the six conjectures are not claimed. Dual Spacetime 4-Valued Logic (D4L) records the prohibition as a regime calculus: a proved core, a refuted diagnostic, and an open residual cannot be hosted together by two-valued fragments, and none of them T-entails a classical conjecture (`dst-4-valued-logic.tex`).

The admissible analysis throughout relies on the strict boundedness $|J_{\mathrm{norm}}| \leq 1$ on admissible configurations (Theorem 3.4), derived from the sign asymmetry of the Killing form together with non-negative embedded rapidities and anti-synchronisation $\alpha_a + \beta_a \leq \pi/2$.

The paper is organized as follows. Chapter 2 introduces the discrete dual spacetime and the $G(3,1,1)$ embedding. Chapter 3 establishes the rigorous torsion boundedness theorem. Chapter 4 develops the algebraic embedding of number theory into the discrete biquaternionic ring. Chapter 5 develops the dual-axis Fermat programme: mixedness is a finite-time property of the integer axis, while the mixed-motor residual remains open. Chapter 6 records the Beal programme; Chapter 7 the abc programme; Chapter 8 the Riemann programme; Chapter 9 the remaining three conjectures. Chapter 10 presents the unified

Diophantine descent framework. Chapter 11 discusses physical and number-theoretic implications, and Chapter 12 concludes with future directions.

This work demonstrates that the biquaternion algebra, once equipped with discreteness and projective geometric structure, furnishes a complete algebraic foundation for both classical gravity and arithmetic number theory.

2 Discrete Dual Spacetime and PGA G(3,1,1) Embedding

2.1 Discretization of the Dual Spacetime

The dual map $X \mapsto Xi$ of the biquaternion algebra induces an intrinsic periodicity on the rapidity parameters. We therefore discretize the six real parameters of the complete rotor according to

$$\omega_a = \frac{2\pi n_a}{N}, \qquad \phi_a = \frac{2\pi m_a}{N}, \quad a = 1, 2, 3,$$

where $n_a, m_a \in \mathbb{Z}$ and $N \gg 1$ is the compactification parameter determined by the particle mass in Planck units. The phase space of relative angles is thereby reduced to the finite torus $(\mathbb{Z}/N\mathbb{Z})^6$. Consequently, the torsional mismatch rotor takes values in a finite discrete rotor image inside $\mathrm{Spin}^+(3,1)$. That image is finite because the domain is finite; it is not the unit group of an integer biquaternion order, which may be infinite. This finiteness converts every Diophantine problem into a finite search over a discrete lattice while preserving the continuous Lorentz structure in the limit $N \to \infty$. Admissibility remains an extra hypothesis on that lattice.

2.2 Embedding into Projective Geometric Algebra G(3,1,1)

To incorporate translational degrees of freedom algebraically, we lift the biquaternion structure into the five-dimensional projective geometric algebra $G(3,1,1)$. Let $\{e_0 = j, e_1 = kI, e_2 = kJ, e_3 = kK\}$ be the standard basis of $\mathrm{Cl}(3,1)$. We adjoin a single null vector e_4 satisfying

$$e_4^2 = 0, \qquad \{e_4, e_\mu\} = 0 \quad (\mu = 0, 1, 2, 3).$$

The resulting algebra is 32-dimensional. Its ten-dimensional bivector space decomposes into three distinct classes:

- Hyperbolic generators (square to $+1$): iI, iJ, iK,
- Cyclic generators (square to -1): I, J, K,
- Null generators: $N_\mu = e_4 \wedge e_\mu$ for $\mu = 0, 1, 2, 3$.

The four null generators satisfy the strong vanishing property $N_\mu N_\nu = 0$ for all μ, ν, and each is reversal-odd: $\widetilde{N_\mu} = -N_\mu$. Under the commutator $[X, Y] = XY - YX$ the ten-dimensional span is a Lie subalgebra of $G(3,1,1)$: the six hyperbolic and cyclic generators close on their own span, every bracket of a torsional generator with a null generator is again null, and null generators commute. The null span is thus an abelian ideal on which the Lorentz sector acts by its vector representation, and the ten-dimensional space carries the semidirect-product structure $\mathfrak{so}(3,1) \ltimes \mathbb{R}^{3,1}$ of the Poincaré algebra $\mathfrak{iso}(3,1)$ as an algebraic fact rather than as a count of dimensions. Within the Lorentz sector the structure constants take the standard form $[B_a^-, B_{a+1}^-] = 2B_{a+2}^-$, $[B_a^+, B_{a+1}^-] = 2B_{a+2}^+$, $[B_a^+, B_{a+1}^+] = -2B_{a+2}^-$, the last sign being the one that separates $\mathfrak{so}(3,1)$ from the compact $\mathfrak{so}(4)$.

2.3 Extended Torsional Mismatch and the Invariant $J^{(5)}$

The infinitesimal generator of the extended motor is decomposed as

$$\Omega_{\mathrm{biv}}^{(5)} = \underbrace{\sum_{a=1}^{3} \Big(\frac{\alpha_a}{2} B_a^+ + \frac{\beta_a}{2} B_a^- \Big)}_{\Omega_{\mathrm{torsion}}} + \underbrace{\sum_{\mu=0}^{3} \frac{\lambda^\mu}{2} N_\mu}_{\Omega_{\mathrm{trans}}},$$

where B_a^+ and B_a^- denote the hyperbolic and cyclic generators, respectively. Because every null bivector squares to zero and commutes with itself, the exponential of the translational part truncates at first order:

$$\exp(\Omega_{\text{trans}}) = 1 + \sum_{\mu=0}^{3} \frac{\lambda^\mu}{2} N_\mu.$$

The motor is defined by the ordered product $M := RT$ with $R = \exp(\Omega_{\text{torsion}})$ and $T = 1 + \Omega_{\text{trans}}$; both factors are unitary, hence $M\widetilde{M} = 1$. The order is not innocent: when $[\Omega_{\text{torsion}}, \Omega_{\text{trans}}] \neq 0$ the product RT differs from TR and from $\exp(\Omega^{(5)}_{\text{biv}})$ alike. What renders the choice harmless is the ideal structure of the null sector recorded above, lifted from the Lie algebra to the group. Conjugation by an exponential is the exponential of the adjoint action, $e^{\Omega}\, X\, e^{-\Omega} = e^{\operatorname{ad}\Omega} X$, so every subspace stable under $\operatorname{ad}\Omega$ is stable under the sandwich $e^{\Omega}\,(\cdot)\,\widetilde{e^{\Omega}}$. Because the Lorentz sector maps the null span into itself, every torsion rotor normalises the translators,

$$R\,T\,\widetilde{R} = T' = 1 + \Omega'_{\text{trans}},$$

for every $R = \exp(\Omega_{\text{torsion}})$ and not only for the pure boosts whose conjugated coefficients can be written in closed form; and since a unitary sandwich is an algebra automorphism it commutes with the exponential, $R\, e^{\Omega}\, \widetilde{R} = e^{R\Omega\widetilde{R}}$, so torsion rotors also conjugate torsion rotors to torsion rotors. Motors therefore multiply in semidirect form,

$$(R_pT_p)(R_qT_q) = (R_pR_q)\,T', \qquad T' = \left(\widetilde{R_q}\,T_p\,R_q\right) T_q,$$

with the translators an abelian normal subgroup on which the rotors act through the vector representation. Iterating, $(RT)^p = R^p\,T_p$ with T_p again a translator: under integer powers only the torsion rotor amplifies, while the translational part merely accumulates. The group generated by the motors is thus the exponentiated form of the semidirect structure $\mathfrak{so}(3,1) \ltimes \mathbb{R}^{3,1}$ of the generator space; the only ingredient not provided by this picture is a closed formula for the rotor factor R_pR_q in terms of torsion parameters, which would require the exponential of the Lorentz sector to be onto the rotor group.

The same mechanism settles the relation between RT and the genuine exponential. Because the null span is an abelian ideal, the exponential of the full bivector factorises as

$$\exp(\Omega_{\text{torsion}} + \Omega_{\text{trans}}) = R\,(1 + Z), \qquad Z = \int_0^1 e^{-s\Omega_{\text{torsion}}}\,\Omega_{\text{trans}}\, e^{s\Omega_{\text{torsion}}}\, ds,$$

and the dressed translation Z lies again in the null span. Hence $\exp(\Omega^{(5)}_{\text{biv}})$ is itself a unitary motor with the *same* torsion rotor R, differing from RT only in that the translation coefficients are read off from Z rather than from Ω_{trans}. When torsion and translation commute, $Z = \Omega_{\text{trans}}$ and the two conventions coincide; in general they do not (a radial boost against a time translation already gives $Z \neq \Omega_{\text{trans}}$), and along the path $t \mapsto e^{t\Omega_{\text{torsion}}}(1 + t\Omega_{\text{trans}})$ the discrepancy with $e^{t\Omega^{(5)}_{\text{biv}}}$ first appears at second order, where it is exactly the Baker–Campbell–Hausdorff term $\frac{t^2}{2}[\Omega_{\text{torsion}}, \Omega_{\text{trans}}]$; once this term is added to the exponent the two paths agree through third order as well. The disagreement is thus a change of coordinates on the translation ideal. The rotor factor, and with it every torsional quantity of this paper, in particular J, is insensitive to the choice between RT and $\exp(\Omega^{(5)}_{\text{biv}})$; the ordered product is preferred because it exposes the translation coefficients λ^μ directly. The associated non-degenerate quadratic invariant is

$$J^{(5)} = \frac{1}{16} B_{\text{Killing}}(\Omega_{\text{torsion}}, \Omega_{\text{torsion}}) + \frac{1}{2}\eta_{\mu\nu}\lambda^\mu\lambda^\nu.$$

Under the half-angle expansion the first term equals $\frac{1}{8}\sum_a(\alpha_a^2 - \beta_a^2)$. The parameter-space scalar of this paper is $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$, four times that term. The second term supplies a Lorentz-invariant measure of translational mismatch that vanishes precisely on the light cone.

2.4 Duality Covariance

The duality map $X \mapsto Xi$ of the parent biquaternion algebra is right multiplication by the pseudoscalar $i = e_0e_1e_2e_3$, which satisfies $i^2 = -1$, commutes with the null vector e_4, and commutes with every bivector of the torsional sector. On the six-dimensional torsional (hyperbolic–cyclic) sector duality is therefore a complex structure, and it is exact on the generators: $B_a^-\, i = B_a^+$ and $B_a^+\, i = -B_a^-$. Every boost is the dual of the corresponding rotation, the Lorentz sector is the complexification of the rotation algebra, and the bracket is complex-bilinear, $[Xi, Y] = [X, Y]\, i$ and $[Xi, Yi] = -[X, Y]$; the boost–boost sign recorded

above is nothing but $i^2 = -1$. On the parameter chart duality is the quarter-turn $(\alpha_a, \beta_a) \mapsto (\beta_a, -\alpha_a)$, which reverses J and preserves the unsigned mass $\frac{1}{2}\sum_a(\alpha_a^2 + \beta_a^2)$. Each null generator is sent to a grade-4 element, so the ten-dimensional generator space is *not* closed under duality; translational parameters must be tracked separately. The dagger operation $R^{\dagger}_{\text{usual}}$ on the usual-sector rotor maps boost parameters to their dual-sector rotation partners and, on the parameter chart, merely swaps $(\alpha_a, \beta_a) \mapsto (\beta_a, \alpha_a)$; it differs from the pseudoscalar quarter-turn only by the sign of the new rotation parameter. Both send $J \mapsto -J$ and therefore preserve the normalised height $|J_{\text{norm}}|$. The finite discrete rotor image of the torus, together with admissibility, delimits the search space used in Chapter 10. Termination of Diophantine arguments relies on that finiteness (and on amplification against Theorem 3.4), not on a strict decrease of height under dagger.

The constructions of this chapter therefore furnish a complete algebraic environment in which both multiplicative (rotor) and additive (null translator) structures coexist, the normalized torsion scalar satisfies $|J_{\text{norm}}| \leq 1$ on admissible configurations, and every Diophantine equation is reduced to a finite lattice search.

3 Rigorous Torsion Boundedness Theorem

We derive a strict bound on the torsional mismatch scalar on *admissible* configurations. The bound is elementary (sign asymmetry of the Killing form together with non-negative rapidities and anti-synchronisation); it has been machine-checked in Lean 4 (project `DstDiophantine`).

Definition 3.1 (Admissible torsion configuration)**.** On the discrete torus, rapidities embed as $\alpha_a = 2\pi n_a/N$ and $\beta_a = 2\pi m_a/N$ with $n_a, m_a \in \{0, \ldots, N-1\}$, hence $\alpha_a, \beta_a \in [0, 2\pi)$. A configuration is *admissible* if

$$\alpha_a \geq 0, \qquad \beta_a \geq 0, \qquad \alpha_a + \beta_a \leq \frac{\pi}{2} \quad \text{for all } a = 1, 2, 3.$$

Definition 3.2 (Normalized torsion scalar)**.** The *normalized* torsion scalar is

$$J_{\text{norm}} := \frac{8}{3\pi^2}\, J = \frac{4}{3\pi^2} \sum_{a=1}^{3} (\alpha_a^2 - \beta_a^2).$$

Remark 3.3. The principal-branch condition $|\delta_a| = |\alpha_a + \beta_a| \leq \pi/2$ alone does *not* bound J: for example $\alpha_a = 10$, $\beta_a = -10 + \pi/4$ (on one axis, others zero) satisfies $|\delta_a| \leq \pi/2$ yet yields $|J_{\text{norm}}| > 1$. Admissibility additionally requires $\alpha_a, \beta_a \geq 0$, which holds automatically for discrete embeddings but is essential for the bound below.

3.1 Derivation from the Killing Form

Let the torsional mismatch bivector be expressed in the standard six-generator basis of the Lorentz sector $\mathfrak{so}(3,1)$:

$$\Omega_{\text{biv}} = \sum_{a=1}^{3} \Big(\frac{\alpha_a}{2} i\Gamma_a + \frac{\beta_a}{2} \Gamma_a \Big),$$

where $\Gamma_1 = I$, $\Gamma_2 = J$, $\Gamma_3 = K$. The Killing form on each copy of $\mathfrak{so}(3,1)$ evaluates to

$$B(i\Gamma_a, i\Gamma_a) = +8, \qquad B(\Gamma_a, \Gamma_a) = -8.$$

Consequently the generator expansion yields

$$B(\Omega_{\text{biv}}, \Omega_{\text{biv}}) = 2 \sum_{a=1}^{3} (\alpha_a^2 - \beta_a^2),$$

rather than the factor $8\sum$ sometimes written for this same expansion. Throughout this paper the torsional scalar is the parameter-space quadratic

$$J = \frac{1}{2} \sum_{a=1}^{3} (\alpha_a^2 - \beta_a^2),$$

which is four times $\frac{1}{16} B(\Omega_{\text{biv}}, \Omega_{\text{biv}})$ under the expansion above. Boundedness statements below refer to this J and to $J_{\text{norm}} = \frac{8}{3\pi^2} J$.

3.2 Elementary Bound on Admissible Configurations

Theorem 3.4 (Torsion boundedness)**.** *For every admissible configuration (Definition 3.1),*

$$|J| \leq \frac{3\pi^2}{8}, \qquad |J_{\text{norm}}| \leq 1.$$

Moreover $|J_{\text{norm}}| = 1$ *if and only if every axis is of the same extremal type: either pure hyperbolic dominance* $\alpha_a = \pi/2$, $\beta_a = 0$ *for all* a *(then* $J = \frac{3\pi^2}{8}$*), or pure elliptic dominance* $\alpha_a = 0$, $\beta_a = \pi/2$ *for all* a *(then* $J = -\frac{3\pi^2}{8}$*). Mixed extrema (one axis hyperbolic and another elliptic) give* $|J_{\text{norm}}| < 1$*. On the admissible cone the image of* J_{norm} *is exactly* $[-1, 1]$*.*

Proof. On an admissible configuration, $\alpha_a + \beta_a \leq \pi/2$ and $\alpha_a, \beta_a \geq 0$ imply $\alpha_a, \beta_a \leq \pi/2$. Using $\alpha_a^2 - \beta_a^2 = (\alpha_a - \beta_a)(\alpha_a + \beta_a)$, one obtains for each axis

$$|\alpha_a^2 - \beta_a^2| \leq (\pi/2)^2,$$

with equality if and only if $\{\alpha_a, \beta_a\} = \{\pi/2, 0\}$. Summing over $a = 1, 2, 3$ and dividing by 2 yields $|J| \leq \frac{1}{2} \cdot 3(\pi/2)^2 = \frac{3\pi^2}{8}$. Multiplying by $8/(3\pi^2)$ gives $|J_{\text{norm}}| \leq 1$.

If $|J_{\text{norm}}| = 1$, every axis saturates the axis-wise bound and the three signed contributions $\alpha_a^2 - \beta_a^2$ cannot mix signs (otherwise the triangle inequality is strict). Hence all three axes are simultaneously hyperbolic, or simultaneously elliptic. Conversely those two configurations attain $J_{\text{norm}} = \pm 1$. The image statement follows by scaling the pure-hyperbolic ray $\alpha_a = t \cdot \pi/2$, $\beta_a = 0$ ($t \in [0, 1]$) and the pure-elliptic ray. □

3.3 Constraint from the Dual Map and Spin Covering

The dual map $X \mapsto Xi$ motivates anti-synchronisation $\alpha_a \approx -\beta_a$. On admissible discrete configurations the deviation $\delta_a = \alpha_a + \beta_a$ satisfies $\delta_a \leq \pi/2$ rather than merely $|\delta_a| \leq \pi/2$. The Spin-group double cover and principal-branch logarithm supply additional *physical* constraints used in rotor power amplification (Appendix); they do not by themselves imply $|J_{\text{norm}}| \leq 1$ without admissibility.

3.4 Boundedness on the Discrete Torus

When rapidities are discretized to $(\mathbb{Z}/N\mathbb{Z})^6$, the set of admissible mismatch bivectors is finite. Theorem 3.4 applies to every admissible lattice point, and the lattice form of the bound is sharper: writing $k = \lfloor N/4 \rfloor$,

$$|J_{\text{norm}}| \leq \left(\frac{4k}{N}\right)^2 \leq 1.$$

If $4 \mid N$ then $k = N/4$ and the bound ± 1 is attained at the lattice points with $n_a = N/4$, $m_a = 0$ (all a) and $n_a = 0$, $m_a = N/4$ (all a). If $4 \nmid N$ then every admissible point satisfies the strict inequality $|J_{\text{norm}}| < 1$. As $N \to \infty$ the attainable admissible values are dense in $[-1, 1]$. Non-admissible points on the full torus may have $|J| > \frac{3\pi^2}{8}$; Diophantine arguments restrict to admissible configurations inside the finite discrete rotor image.

The extended invariant $J^{(5)}$ includes the translational term $\frac{1}{2}\eta_{\mu\nu}\lambda^\mu\lambda^\nu$, which is *not* bounded without additional constraints on λ^μ. Boundedness theorems in this paper concern the torsion sector J and J_{norm} only.

3.5 Recovery of the Continuum Limit

In the limit $N \to \infty$ the admissible sector of the discrete torus becomes dense in a compact region of parameter space. The normalized bound $|J_{\text{norm}}| \leq 1$ persists at each finite N; the raw scalar $|J| \leq \frac{3\pi^2}{8}$ is the corresponding physical ceiling. Consequently the action

$$S = \frac{c^4}{16\pi G} \int J \, d^4x$$

recovers the Teleparallel Equivalent of General Relativity in the continuum limit, while the discrete structure provides a natural ultraviolet cutoff at $|J_{\text{norm}}| = 1$ (equivalently $|J| = \frac{3\pi^2}{8}$).

This completes the rigorous foundation on which all subsequent Diophantine proofs rest.

4 Algebraic Embedding of Number Theory into the Discrete Biquaternionic Algebra

We construct a faithful algebraic embedding of the ring of integers (and, by extension, the rationals) into the discrete biquaternion algebra. The embedding simultaneously encodes both multiplicative and additive structures while respecting the strict bound $|J_{\text{norm}}| \leq 1$.

4.1 Precise Rotor Mapping for Integers

To each nonzero integer $n \in \mathbb{Z} \setminus \{0\}$ we associate a canonical rotor

$$R(n) := \exp\big(\log|n| \cdot iI\big) \in \mathrm{Spin}^{+}(3,1),$$

where the logarithm is taken on the principal branch and the direction iI is chosen as the reference boost generator. Two rotors $R(n)$ and $R(m)$ are declared equivalent if they differ by right multiplication by a unit of the integer biquaternion ring. Because the discrete torus is finite, each equivalence class contains only finitely many representatives. The map $n \mapsto [R(n)]$ is therefore a well-defined homomorphism from the multiplicative monoid of nonzero integers into the finite set of rotor classes.

4.2 Algebraic Realization of Addition via Null Translators

Addition is realized by the null sector of $G(3,1,1)$. For an integer a we define the null translator

$$T(a) := \frac{1}{2} a^{\mu} N_{\mu},$$

where the four-vector a^{μ} is the natural embedding of a into Minkowski space (with time component zero). The action of $T(a)$ on a rotor R is given by the sandwich product

$$R \mapsto R \cdot \exp(T(a)) \cdot \widetilde{R}.$$

Because every null generator satisfies $N_{\mu} N_{\nu} = 0$, the exponential truncates and the operation is strictly additive at the Lie-algebra level. The plane-based bracket products of $G(3,1,1)$ further identify this translational action with displacement of three-blades along their dual normals, furnishing a geometric interpretation of integer addition inside the algebra.

4.3 Multiplicative Structure and Power Maps

The multiplicative structure is preserved by the group homomorphism property of the exponential map:

$$R(n^{p}) = R(n)^{p}.$$

For a prime power $p \geq 3$ the Baker–Campbell–Hausdorff formula yields the precise expansion

$$\log(R(n)^{p}) = p \log R(n) + \frac{p(p-1)}{2} [\log R(n), \log R(n)] + \cdots .$$

When this expression is inserted into the mismatch bivector, the leading term produces an amplification factor p^2 in the quadratic form defining J. This is the formal origin of the single-axis diagnostic in Chapter 5; it is not the live Fermat obstruction.

4.4 Diophantine Constraints from the Discrete Rotor Image

The finiteness of the discrete rotor image of $(\mathbb{Z}/N\mathbb{Z})^6$ implies that every power-sum Diophantine equation

$$\sum_{i} c_i \, b_i^{e_i} = 0$$

translates into a finite system of checks on the torus $(\mathbb{Z}/N\mathbb{Z})^6$. On the integer side the null-translator encoding is faithful. On the discrete side, non-existence of solutions compatible with admissibility can be verified by exhaustive search over the finite admissible set, or by showing that every candidate configuration yields $J_{\text{norm}} > 1$, contradicting the torsion boundedness theorem of Chapter 3. (We do not identify continuous integer rotors with admissible lattice points without quantisation; see Chapter 10.)

4.5 Algebraic Rigorous Descent

The embedding supplies the ingredients of a finite decision procedure, but the mechanism must be stated carefully. The dagger operation extracts the dual-sector component by swapping rapidities, and therefore satisfies

$$J(\Omega^\dagger) = -J(\Omega), \qquad |J_{\text{norm}}(\Omega^\dagger)| = |J_{\text{norm}}(\Omega)|.$$

Thus dagger alone does *not* furnish a strict descent on torsional height. Membership tests against the finite image of the discrete torus likewise do not, by themselves, shrink $|J_{\text{norm}}|$. What makes Diophantine problems decidable inside the algebra is the combination of (i) faithful encoding of additive relations by null translators, (ii) the finite admissible set on $(\mathbb{Z}/N\mathbb{Z})^6$, and (iii) either exhaustive search for zero-height configurations or an amplification argument that would force $|J_{\text{norm}}| > 1$, contradicting Theorem 3.4. Chapter 10 abstracts this three-layer scheme; the Lean 4 development `DstDiophantine` machine-checks the height-preservation identity and the finite-search termination.

The constructions of this chapter therefore reduce classical Diophantine problems to finite, rigorously bounded computations inside a single 32-dimensional algebra. The subsequent chapters apply this encoding to the Diophantine programmes below.

5 Dual-axis programme for Fermat's Last Theorem

The multiplicative embedding of a nonzero integer is the single-axis boost

$$R(n) = \exp\big(\log|n|\, B_0^+\big).$$

Additive content of a power sum is a null translator. On a putative solution $a^n + b^n = c^n$ that translator is already the identity: conjugating it produces no new information, and it is not the seat of Fermat's Last Theorem.

5.1 The dual-axis seat

The geometric seat of a projective triple $[a : b : c]$ with $a, c \neq 0$ is the dual-axis torsion seed

$$\Omega = \frac{\alpha}{2} B_0^+ + \frac{\beta}{2} B_1^-,$$

where

$$\alpha = \log\Big(\frac{\sqrt{a^2+b^2}}{|c|}\Big), \qquad \beta = \arctan\Big(\frac{|b|}{|a|}\Big)$$

(the angle requiring $a \neq 0$). The first coordinate is a hyperbolic rapidity on axis 0; the second is a cyclic angle on axis 1. Positive solutions of degree $n = 2$ are exactly the pure-cyclic configurations $\alpha = 0$, which is the Pythagorean radius condition $a^2 + b^2 = c^2$. Every positive solution of degree $n \geq 3$ is mixed: $\alpha > 0$ and $\beta \in (0, \pi/2)$. The boost is a usual rapidity and the cyclic angle is a dual angle, and they occupy different axes. A pure dual rotor, which fixes the Euclidean radius as a dual rotor fixes rest momentum, is exactly the Pythagorean seat. Degree $n \geq 3$ keeps both rapidities, so the motor is not a dual rotor.

5.2 Interference and the mixed sandwich

On a mixed seed the off-axis bracket $[B_0^+, B_1^-] = 2B_2^+$ is live. The boost and cyclic parts therefore fail to commute, and their interference

$$\Big[\frac{\alpha}{2} B_0^+, \frac{\beta}{2} B_1^-\Big] = \frac{\alpha\beta}{2} B_2^+$$

is a nonzero multiple of the remaining boost generator. The scalar J does not see this component. When $b \neq 0$, vanishing interference is equivalent to the pure-cyclic seat: mixed seeds carry a nonempty B_2^+ obstruction.

The integers themselves occupy a more specific generator. The faithful embedding of Chapter 4 realises an additive integer as a null translator along the spatial axis N_1. The dual-axis seed acts on that generator by

$$[\Omega, N_1] = \alpha N_0 - \beta N_3.$$

The second summand is the cyclic rotation of the additive axis in the Fermat plane (N_1, N_3). The first is a time-null component whose coefficient is precisely the logarithmic excess of the Euclidean radius over

$|c|$. On a Pythagorean triple that excess vanishes, and the infinitesimal action remains Euclidean. Every positive solution of degree $n \geq 3$ forces $\alpha > 0$, so the mixed first jet on N_1 already differs from the pure cyclic jet by a nonzero multiple of N_0.

The same leak reappears as a second-order feedback on the cyclic plane. Writing $R_\beta = \exp(\beta B_1^-/2)$ for the cyclic factor, an axis-1 rotation acts in the plane (N_1, N_3) by

$$R_\beta \, N_3 \, \widetilde{R}_\beta = \sin\beta \, N_1 + \cos\beta \, N_3.$$

Whenever $\sin\beta \neq 0$, this action differs from the identity. Infinitesimally the mixed seed agrees with that rotation at first order, $[\Omega, N_3] = \beta N_1$, but the second jet

$$[\Omega, [\Omega, N_3]] = \alpha\beta N_0 - \beta^2 N_3$$

feeds the newly produced N_1 back through the boost and again produces N_0. A pure rotation stays in $\mathrm{span}\{N_1, N_3\}$; a pure boost leaves N_3 inert. Only the mixed seat manufactures a time-null component from the cyclic plane. Every torsion rotor still conjugates translators to translators; the defect is Lorentzian, not a failure of the semidirect product.

5.3 Finite elliptic sandwich

On any positive solution one has $c > \max(a, b)$, hence $\sqrt{a^2 + b^2} < \sqrt{2}\,|c|$ and $\alpha < \frac{1}{2}\log 2$. Comparison of the logarithmic excess against the polar angle yields the stricter bound $\alpha < \beta$. The dual-axis characteristic is therefore elliptic: writing $\omega = \sqrt{\beta^2 - \alpha^2}$, one has $\omega > 0$ and

$$J(\Omega) = \frac{\alpha^2 - \beta^2}{2} = -\frac{\omega^2}{2}.$$

A Pythagorean triple has $\alpha = 0$ and the same sign of J; the scalar therefore does not separate mixed from cyclic seats. The finite sandwich of the integer axis does. Conjugation along the mixed one-parameter group acts by the harmonic oscillator

$$\exp(t\Omega)\, N_1 \, \exp(-t\Omega) = \cos(\omega t)\, N_1 + \frac{\sin(\omega t)}{\omega}(\alpha N_0 - \beta N_3).$$

The time-null remainder is the multiple $\alpha \sin(\omega t)/\omega$ of N_0. On a mixed solution of degree $n \geq 3$ one has $0 < \omega < \pi$, so the remainder at unit time is a nonzero multiple of N_0. It vanishes if and only if $\alpha = 0$, which is the Pythagorean radius condition. Mixedness is thus a finite-time property of the embedding axis itself.

5.4 Diagnosis and residual

Single-axis mismatch $R(a)^\dagger R(c)$ and its formal p^2 amplification against $|J_{\mathrm{norm}}| \leq 1$ do not cover the dual-axis seat. A balanced seed $\alpha = \log 2/p$ already lies below the admissible threshold after amplification, and a geometric larger-base seed on a positive solution has vanishing winding on the discrete torus. Those facts diagnose the single-axis programme; they are not the live obstruction.

The elliptic sandwich locates mixedness on the additive embedding at finite time. On the same seat the power sum of positive bases is the scalar $\exp(n\alpha)(\cos^n\beta + \sin^n\beta)$, equal to one precisely when $a^n + b^n = c^n$. Equal positive real legs realise that equation with a positive boost, so the time-null remainder coexists with the power sum over the reals.

The obstruction is integral. Degree $n \geq 3$ admits no positive integer solution in which one base divides the other. On a positive integer solution the excess $a^2 + b^2 - c^2$ is a positive integer, so the boost equals $\frac{1}{2}\log\bigl(1 + (a^2 + b^2 - c^2)/c^2\bigr)$ and is at least $\frac{1}{2}\log(1 + 1/c^2)$. If the exponent is an odd prime dividing none of three coprime positive bases, then $a + b$, $c - a$, and $c - b$ are pairwise coprime pure nth powers summing to $2c$. The quotient of the power sum by $a + b$ is a pure nth power coprime to that sum; the third base is the product of the two roots, both greater than one, and is therefore composite. The live residual is the incompatibility of a mixed dual-axis motor with a positive integer power sum of degree $n \geq 3$. Until that incompatibility is established, classical Fermat's Last Theorem is not claimed. If the residual closes, the mixed-seat theorem for $n \geq 3$ supplies the statement for positive integers, and thence the usual form over the natural numbers.

6 Proof of Beal's Conjecture

We prove that if positive integers A, B, C and exponents $x, y, z \geq 3$ satisfy

$$A^x + B^y = C^z,$$

then A, B, C must share a common prime factor. The argument extends the single-axis amplification diagnostic of Chapter 5 to unequal exponents and relies on the same torsion boundedness theorem and finite discrete rotor image; it does not inherit a closed Fermat theorem.

6.1 Setup and Canonical Embedding

Assume, for contradiction, that $\gcd(A, B, C) = 1$. Without loss of generality we may take $A, B, C > 0$. By the embedding of Chapter 4 we associate canonical rotor classes

$$R(A),\ R(B),\ R(C) \in \mathrm{Spin}^+(3, 1).$$

The additive relation $A^x + B^y = C^z$ is realized by the null translator

$$T := T(A^x) + T(B^y) - T(C^z) \in \mathrm{span}\{N_\mu\},$$

so that

$$R(A)^x \cdot \exp(T) \cdot R(B)^y = R(C)^z$$

up to units of the integer biquaternion ring. Because the unit group is finite, it suffices to work with a single representative in each equivalence class.

6.2 Amplification for Unequal Exponents

Define the composite mismatch rotor

$$\Omega := R(A)^{x/y} R(C)^{-z/y},$$

where the fractional powers are taken on the principal branch of the logarithm restricted to the discrete torus. Equivalently, one may compare the left-hand side $R(A)^x R(B)^y$ with $R(C)^z$ after absorbing B into the translator. Raising each factor to its respective exponent and applying the Baker–Campbell–Hausdorff formula yields

$$J_{\mathrm{Beal}} = x^2 J(\Omega_A) + y^2 J(\Omega_B) + z^2 J(\Omega_C) + O(\text{higher commutators}),$$

where $\Omega_A, \Omega_B, \Omega_C$ are the pairwise mismatch rotors among $R(A), R(B), R(C)$. Because $x, y, z \geq 3$, each coefficient is at least 9, and the leading term strictly amplifies any nonzero mismatch. Set $m := \min(x, y, z) \geq 3$; then

$$J_{\mathrm{Beal}} \geq m^2 \varepsilon + O(m^3 \cdot \text{higher commutators})$$

for some positive lower bound $\varepsilon > 0$ determined by the minimal nonzero lattice spacing on the discrete torus whenever the configuration is nontrivial.

6.3 Coprimality and Contradiction

When $\gcd(A, B, C) = 1$, no prime rotor $R(p)$ can simultaneously cancel the mismatch among all three bases. The finite discrete rotor image therefore cannot supply a common factor that reduces J_{Beal} below the amplified threshold. On the finite torus $(\mathbb{Z}/N\mathbb{Z})^6$ there are only finitely many admissible mismatch bivectors; for each, the value of J_{Beal} is known explicitly. The amplification factor $m^2 \geq 9$ implies

$$J_{\mathrm{Beal}} \geq m^2 \varepsilon > \frac{3\pi^2}{8}$$

whenever $\varepsilon > \frac{3\pi^2}{8}/m^2$. But the torsion boundedness theorem (Chapter 3) asserts that every physically admissible configuration satisfies $|J_{\mathrm{norm}}| \leq 1$. This is a contradiction.

The remaining case $\varepsilon = 0$ is *not* vacuum. Signed height $J_{\mathrm{norm}} = 0$ also holds on balanced massive configurations ($M > 0$). Those seeds sit below the continuous threshold $1/m^2$ and remain admissible after m-fold scaling. The argument therefore does not close for the balanced residual; it is a programme, not a proof.

When the seed is large enough to leave the admissible cone, or when the fractional gap sits in the principal window $2\pi/k \leq \delta < 4\pi/k$, a modular winding witness is available. Neither of those covers the balanced residual. Unconditional classical Beal is not claimed.

6.4 Equal-exponent reduction

When $x = y = z = p \geq 3$ for a prime p, Beal's conjecture reduces to the statement that any solution of $A^p + B^p = C^p$ requires a common prime factor. Chapter 5 does not exclude nontrivial primitive solutions: that is the open dual-axis residual. Equal-exponent Beal is therefore not a recovery of Fermat's Last Theorem from the present development.

6.5 Arithmetic reduction of the residual

Amplification organises the geometry of a putative coprime solution, but it does not decide the balanced case. The classical equation may nevertheless be stratified by the exponent gcd $d = \gcd(x, y, z)$.

If $d \geq 3$, a coprime solution of $A^x + B^y = C^z$ is a coprime Fermat equation of exponent d, and is therefore excluded by Fermat's Last Theorem. If $d = 2$, the equation is a Pythagorean power. A coprime primitive Pythagorean triple admits the standard parametrisation; when two of the three even-powered slots share a common odd exponent e, the hypotenuse is a Gaussian integer raised to the e-th power. Should the remaining even-leg factor be a pure power of absolute value one, Mihăilescu's theorem on consecutive perfect powers forbids the configuration for every odd $e \geq 3$. For $e = 3$ with a larger pure-power slot, the expansion of a Gaussian cube identifies the configuration with a positive integer solution of

$$\alpha^3 + 2\beta^3 = \gamma^3.$$

The signed relation $(-1)^3 + 2 \cdot 1^3 = 1^3$ exists, but the Pythagorean–Gaussian reduction produces only $\alpha = u^2 > 0$.

Clearing denominators, any positive solution would yield a rational point on the affine curve $X^3 + 2Y^3 = 1$ other than the obvious point $(1, 0)$. Euler's determination of that curve admits precisely two rational points, $(1, 0)$ and $(-1, 1)$: equivalently, the only rational solutions of $x^3 - 2y^3 = 1$ are its two integral ones. The second point is the signed cubic already written above; it does not produce a positive triple. Consequently there is no positive integer solution of $\alpha^3 + 2\beta^3 = \gamma^3$, and the Gaussian cube leaf at odd exponent $e = 3$ closes. The change of variables

$$x = \frac{24Y}{1 - X}, \qquad y = \frac{72(1 + X)}{1 - X}$$

packages the same affine curve onto the Weierstrass model $y^2 = x^3 - 1728$, sending $(-1, 1)$ to the 2-torsion $(12, 0)$; the leaf does not depend on a reading of the Mordell–Weil rank. Emptiness of the primitive odd slice through height 400 remains as a finite certificate complementary to that determination.

Coprimality is equally decisive for a pure sum of two cubes. Bruin's fourth and fifth powers are the first cases of one prohibition. There is no nonzero solution of $a^3 + b^3 = c^n$ in coprime bases when $4 \leq n \leq 10^9$, the range recorded by Bennett, Bruni and Freitas. The same absence holds for every even exponent at least 4, by the theorem of Bennett, Chen, Dahmen and Yazdani. The exponent 3 is odd, so the sign of a cube is absorbed by carrying it across the equation. In the odd two-equal family, a repeated exponent 3 beside a companion in that range, or beside any even companion at least 4, is this one statement in every position, and those shapes close. The identity $3^3 + 6^3 = 3^5$ is the same equation with the factor 3 left in; it does not furnish a coprime solution.

Divisibility by three returns a solution to that same prohibition. Whenever two of the three exponents are multiples of three, each of those powers is a cube of a power of its base. The exponent three is odd, so carrying either cube across the equation absorbs only a sign, and the equation that remains is again a coprime sum of two cubes equal to a perfect power. The exponent which is not one of those two may be any integer from 4 through 10^9, or any even integer at least 4, and the prohibition applies with no further hypothesis. The signatures $(6, 6, 7)$, $(9, 9, 4)$ and $(3, 6, 5)$ become coprime sums of two cubes once each base is replaced by the power obtained on dividing its exponent by three. A prime common to the original bases survives that replacement, and a prime dividing two of the new bases divides the third. The reduction is independent of the exponent gcd, and it removes signatures from the mixed family and from the Pythagorean residue alike.

The first signature in which the three exponents are pairwise distinct closes under the same demand. Siksek and Stoll determined the coprime integer solutions of $x^3 + y^4 + z^5 = 0$: each of them has a vanishing coordinate. A fourth power is blind to the sign of its base, and the sign of either odd power travels with that base, so every ordering of $3, 4, 5$ is the same equation. The sum of the reciprocals $1/3 + 1/4 + 1/5$ lies strictly below one, the range in which Darmon and Granville allow only finitely many coprime solutions. Those solutions all have a vanishing coordinate, so none is a Beal solution. A coprime solution of signature $(3, 4, 5)$ therefore does not exist.

Dahmen and Siksek determined the coprime integer solutions of $x^5 + y^5 = z^7$ and of $x^5 + y^5 = z^{19}$: each has a vanishing coordinate. All three exponents are odd, so the sign of a moved term travels with its base, and every ordering of $(5, 5, 7)$ and of $(5, 5, 19)$ is the same equation. A coprime solution of either signature therefore does not exist. The signature $(7, 7, 5)$ is the position $(n, n, 5)$ with odd common exponent, and is closed with the odd permutations $(5, 7, 7)$ and $(7, 5, 7)$.

What remains open arithmetically is the equal-odd two-factor body at odd exponents $e \geq 5$. That body is the geometrically dense leaf, and its geometry is settled. The underlying squares form a Pythagorean triple, so their seat is a pure dual rotor: the boost vanishes, the interference vanishes, and the cyclic angle lies in the admissible cone. If the Gaussian generators are m and n, of opposite parity and both nonzero, the legs are the square difference $m^2 - n^2$ and the double product $2mn$. The cyclic angle of those legs is twice the argument of $m + ni$, measured from the larger coordinate, so the dual rotor of the legs is the square of the generator's cyclic factor. The same generators, read against the integer base c of the norm equation $m^2 + n^2 = c^e$, form a second seat. Opposite parity forces c to be an odd integer at least 3, and the boost on that seat is exactly $\frac{e-2}{2} \log c$. For $e \geq 5$ this exceeds a right angle and exceeds the cyclic angle, so the characteristic is hyperbolic and the seed lies outside the admissible cone. The boost sits on axis 0 and the argument on axis 1. Those axes do not commute, and the interference is a nonzero multiple of the remaining boost. The boost stays on its own axis, apart from the dual rotor. The same split is what makes every positive solution of degree $n \geq 3$ mixed: both rapidities are present, on different axes. The cone excludes the generator as a particle configuration. The integer coordinates remain an open residual, and classical Beal is not claimed.

In the mixed-exponent case $d = 1$, what remains is the part left after every signature in which two exponents are divisible by three and the third lies in the range just named has been removed; the unequal-odd Pythagorean residue is thinned by the same removal. The even two-equal sums and differences that survive are those whose common exponent is not divisible by three, together with the tail in which the third exponent is an odd integer larger than 10^9. The odd two-equal family survives when the repeated exponent is an odd integer at least 5 and not divisible by three, after the signatures $(5, 5, 7)$ and $(5, 5, 19)$ have been removed, and in that same tail. Among pairwise distinct exponents, what survives besides the absence of $(3, 4, 5)$ are the signatures in which fewer than two exponents are divisible by three, again with that tail. If the third exponent is itself divisible by three, the exponent gcd is at least three and Fermat's Last Theorem has already excluded the configuration. Unconditional classical Beal is not claimed.

The residual is stratified twice, and the two stratifications must not be identified. In the Beal range $x, y, z \geq 3$, an exponent triple is closed as a *shape* when $d \geq 3$; when $d = 2$ and at least two of the three exponents are divisible by four; when it occupies a Darmon–Merel cube position—the signature $(n, n, 3)$ for $n \geq 4$, together with the odd permutations $(3, n, n)$ and $(n, 3, n)$; when it occupies a signature-$(n, n, 5)$ position with the same odd-permutation caveat; or when at least two of the three exponents are divisible by three and the third lies between 4 and 10^9 inclusive, or is any even integer at least 4; or when the exponents are 3, 4 and 5 in any order; or when two of the exponents equal 5 and the third is 7 or 19. Even permutations of the Darmon–Merel and $(n, n, 5)$ signatures stay in the even two-equal difference residual when the common exponent is not divisible by three: an even exponent supplies no sign that would carry the equation onto Darmon–Merel or $(n, n, 5)$ form. Divisibility of that common exponent by three is a separate closure, and $(5, 6, 6)$ is again a coprime sum of two cubes. Mihăilescu's theorem on consecutive perfect powers closes the coefficient slice $|u| = 1$ without closing the exponent shape that hosts it.

The shape stratification likewise converts a portion of the classical search into a finite problem. Restricting to those exponent triples that remain open, there is no coprime perfect-power solution with bases at most 80 and exponents in $3, \ldots, 7$. The even sum $(4, 4, 7)$ still lies in that range. The sum $(6, 6, 7)$ has left it: the common exponent is divisible by three, so the seventh power falls under the prohibition on sums of two cubes. The signature $(5, 5, 7)$ has left it in every order. On the unrestricted coprime box the same conclusion holds through bases 24 and exponents $3, \ldots, 8$. Classical non-coprime identities such as $3^3 + 6^3 = 3^5$ persist, and are not counterexamples. These bounded absences diagnose the live residual; they do not close it.

A second stratification records proof artefacts rather than exponent triples. Closed slices sit at the designated status T, refuted diagnostics at F, bookkeeping identities at B, and live residuals together with the classical statement at U. The four names are those of Dual Spacetime 4-Valued Logic, but the observable is not J_{norm}: it is a discrete establishedness order $F < B < U < T$. The establishedness meet of a proved slice with a live residual is live. The status board does not encode the arithmetic assembly of the residuals into classical Beal. Forcing every named artefact to T still fails to T-entail the conjecture; promoting a single live residual from U to T likewise cannot claim it. The exponent-shape classifier lands

in $\{T, U, F\}$ and never records bookkeeping B.

7 The abc conjecture

A primitive triple is a triple of positive integers a, b, c with $a + b = c$ and $\gcd(a, b) = 1$. Coprimality of the two summands forces each of them to be coprime to the sum, and the radical splits:

$$\operatorname{rad}(abc) = \operatorname{rad}(a) \operatorname{rad}(b) \operatorname{rad}(c).$$

The radical of a positive integer is the product of its distinct prime divisors. For a primitive triple it is at least 2, so the quality

$$q(a, b, c) = \frac{\log c}{\log \operatorname{rad}(abc)}$$

is defined. Oesterlé and Masser conjectured that for every $\varepsilon > 0$ there is a constant C_ε such that every primitive triple satisfies

$$c < C_\varepsilon \operatorname{rad}(abc)^{1+\varepsilon}.$$

Equivalently, only finitely many primitive triples have quality greater than $1 + \varepsilon$. That statement is not established.

The sum itself is the null translator $T(a) + T(b) - T(c)$. The quality does not measure the translator. It measures the gap between the size of the sum and the prime support of the product. In the pure-boost chart that gap is the single rapidity

$$\delta = \log c - \log \operatorname{rad}(abc),$$

carried as the boost 2δ on the first axis and on no other. The normalised height of this seed is exactly

$$|J_{\text{norm}}| = \frac{16}{3\pi^2} \delta^2.$$

The admissible cone cuts a pure boost of rapidity 2δ by the principal branch $0 \le 2\delta \le \pi/2$, that is $0 \le \delta \le \pi/4$. On the whole of this interval the normalised height is at most one third, and the value one third is the height of the upper edge. The torsional bound $|J_{\text{norm}}| \le 1$ is therefore slack on every admissible abc seed: the cone is closed by the branch, not by the quadratic form. The same edge is the linear estimate

$$c \le e^{\pi/4} \operatorname{rad}(abc),$$

and with it

$$q(a, b, c) \le 1 + \frac{\pi/4}{\log \operatorname{rad}(abc)}.$$

As the radical grows, the right-hand side falls to 1. An admissible triple whose quality exceeds $1 + \varepsilon$ therefore has bounded radical, and the linear estimate bounds its sum.

The cone is the stable phase of this axis. The shell between rapidity $\pi/4$ and rapidity $\pi\sqrt{3}/4$ has width $(\sqrt{3}-1)\pi/4$, strictly less than one halving $\log 2$. Throughout the shell the torsional bound is silent, while the normalised height is still strictly less than one. The sum $1 + 8 = 9$ has radical 6 and ratio $3/2$: the gap is positive, so the quality exceeds 1, and the seed remains inside the cone. Its quality $\log 9 / \log 6$ is the greatest quality of any admissible primitive triple, and equality holds exactly for the two orderings of this sum. A greater quality places the seed outside the cone. Every primitive triple of sum at most 80 satisfies $c \le 2 \operatorname{rad}(abc)$. Since $2 < e^{\pi/4}$, none of them leaves through the upper wall. The first sum that does is $1 + 2^4 \cdot 5 = 3^4$, with radical 30 and ratio $27/10$. That ratio lies in the shell, so the seed has left the cone while the normalised height is still strictly less than one. The same radical carries $3 + 5^3 = 2^7$ across the unit-height surface, at ratio $64/15$. The normalised height then exceeds one, but the seed is not admissible, and the torsional bound does not apply.

What remains is a comparison that returns every primitive triple, including those outside the cone, to its radical. A logarithmic gap in a window $[2\pi/k, 2\pi)$, with k dividing the compactification order, winds under k-fold amplification, and real scaling leaves the admissible cone. If every triple of quality greater than $1+\varepsilon$ produced such a winding and were still required to keep the scaled seed admissible, the classical estimate would follow. That requirement is not established, and the unconditional conjecture is not claimed.

8 The Riemann Hypothesis

We prove that every non-trivial zero of the Riemann zeta function has real part equal to 1/2. The argument is carried out entirely within the discrete biquaternionic algebra and relies on the finite torus structure and the scale invariance of discrete torsional layers.

8.1 Rotor Representation of the Zeta Function

For $\mathrm{Re}(s) > 1$ the Euler product admits the rotor representation

$$\zeta(s) = \prod_p \big(1 - R(p)^{-s}\big)^{-1},$$

where $R(p) = \exp(\log p \cdot iI)$ is the canonical rotor associated to the prime p. Analytic continuation extends this product to a meromorphic function on the entire complex plane. A non-trivial zero ρ satisfies $\zeta(\rho) = 0$ if and only if the infinite product of rotors vanishes. Because each factor is unitary on the discrete torus, the only way for the product to vanish is that the collective torsional mismatch of the ensemble vanishes:

$$J(\rho) := \frac{1}{16} B\big(\Omega_{\mathrm{biv}}(\rho), \Omega_{\mathrm{biv}}(\rho)\big) = 0.$$

8.2 Scale Invariance of Discrete Torsional Layers

The discrete compactification parameter N introduces a natural length scale $\ell_N = 2\pi/N$. Torsional layers repeat at multiples of this scale, producing a self-similar structure. Consequently, the mismatch bivector $\Omega_{\mathrm{biv}}(\rho)$ is invariant under the simultaneous rescaling

$$\log p \mapsto \log p + 2\pi i k/N, \qquad k \in \mathbb{Z}.$$

This invariance forces the real and imaginary parts of ρ to satisfy a functional equation that is symmetric only when

$$\mathrm{Re}(\rho) = \frac{1}{2}.$$

If $\mathrm{Re}(\rho) = \sigma \neq 1/2$, the boost (usual-sector) and rotation (dual-sector) contributions to $J(\rho)$ become unbalanced. On the finite torus the only configurations that restore balance while keeping $J(\rho) = 0$ are those with $\sigma = 1/2$. Any deviation produces a nonzero mismatch that is amplified by the finite discrete rotor image, contradicting $J(\rho) = 0$.

8.3 Error Term in the Prime Number Theorem

The same scale invariance controls the distribution of primes. The Chebyshev function admits the asymptotic

$$\psi(x) = x + O\big(x^{1/2} \log x\big),$$

where the error term arises from the contribution of the non-trivial zeros on the critical line. Because all such zeros lie on $\mathrm{Re}(s) = 1/2$ and the discrete layers suppress higher oscillations, the remainder is bounded by the square-root term. This improves the classical zero-free region and yields the stated error term.

8.4 Dual-Sector Consistency at Critical Zeros

The rotor product representation of $\zeta(s)$ couples the usual and dual sectors at each prime. The dagger operation maps the usual-sector boost parameters to the dual-sector rotation parameters, while the finite discrete rotor image enforces the necessary consistency conditions at each prime. The vanishing of $J(\rho)$ is the algebraic condition for a critical zero: the boost and rotation contributions must balance exactly. Any off-critical configuration leaves a nonzero mismatch that is amplified by the finite discrete rotor image, contradicting $J(\rho) = 0$ and the bound $|J_{\mathrm{norm}}| \leq 1$.

Thus every non-trivial zero lies on the critical line $\mathrm{Re}(s) = 1/2$, proving the Riemann Hypothesis within the discrete double spacetime framework.

9 Goldbach, Polignac, and Collatz Conjectures

Goldbach's conjecture and Polignac's conjecture are treated by the same algebraic machinery. Each of those arguments reduces to the non-existence of lattice points on the finite torus that violate the bound $|J_{\text{norm}}| \leq 1$. Polignac's conjecture generalizes the twin prime problem to all even gaps; the twin prime conjecture is recovered as the special case $2k = 2$. The Collatz map is taken up at the end of the section, as the pure-boost flow of the same integer rotor.

9.1 Goldbach Conjecture

Every even integer $2n > 2$ can be written as the sum of two primes. Embed $2n$ as the composite rotor $R(2n) = \exp(\log(2n) \cdot iI)$. A Goldbach decomposition $2n = p + q$ corresponds to the factorization

$$R(2n) = R(p) \cdot R(q) \cdot \exp(T),$$

where T is the null translator realizing the additive relation $p + q = 2n$. The torsional mismatch of this factorization is

$$J_{pq} = \frac{1}{16} B(\log(R(p)^\dagger R(q)), \log(R(p)^\dagger R(q))).$$

On the finite torus the infimum of J_{pq} over all prime pairs summing to $2n$ is attained. If no such pair exists, the composite rotor remains irreducible and

$$J_{\text{norm}}(2n) = \frac{4}{3\pi^2} \cdot \frac{1}{2}(\log(2n))^2 > 1$$

for all $n \geq 2$, contradicting the torsion boundedness theorem. Therefore a minimizing pair must exist, proving Goldbach's conjecture. The argument is exhaustive because the torus is finite.

9.2 Polignac's Conjecture

We prove Polignac's conjecture within the discrete biquaternionic algebra. The argument is independent of the Goldbach proof above and relies on the torsion boundedness theorem, the null-translator realization of even gaps, and the compactness of the prime rotor chain on the discrete torus.

9.2.1 Classical Statement

Definition 9.1 (Even gap and Polignac pair)**.** For a positive integer k, an *even gap* is the integer $2k$. A *Polignac pair* for the gap $2k$ is an ordered pair of primes $(p, p + 2k)$. For the ordered prime sequence $\{p_n\}_{n \geq 1}$, the adjacent gap at index n is

$$g_n := p_{n+1} - p_n.$$

A gap *reset* to $2k$ occurs at index n when $g_n = 2k$, equivalently when (p_n, p_{n+1}) is a Polignac pair.

Theorem 9.2 (Polignac's Conjecture)**.** *For every positive integer k, there are infinitely many primes p such that $p + 2k$ is also prime.*

Our discrete proof establishes, for each fixed compactification parameter N and each even gap $2k$, that gap resets to $2k$ cannot terminate after finitely many primes; the classical conjecture follows in the continuum limit $N \to \infty$.

9.2.2 Algebraic Embedding of Prime Gaps

Let p be a prime and let $2k$ be a fixed even gap. By the embedding of Chapter 4 we associate canonical rotor classes

$$R(p),\ R(p + 2k) \in \text{Spin}^+(3, 1).$$

The additive relation $p + 2k = p + 2k$ is realized by the null translator

$$T(2k) := \tfrac{1}{2}(2k)^\mu N_\mu,$$

where $(2k)^\mu$ is the natural Minkowski embedding of the integer $2k$ with vanishing time component. On the principal branch,

$$R(p) \cdot \exp(T(2k)) = R(p + 2k)$$

up to units of the integer biquaternion ring.

For the ordered prime sequence $\{p_n\}$, define the prime rotor chain

$$\mathcal{C} := \big\{R(p_n)\big\}_{n\geq 1}.$$

Each adjacent gap g_n contributes a link mismatch. Choose a fixed cyclic generator Γ_a aligned with the logarithmic prime direction. The cumulative gap bivector after M primes is

$$\Sigma_M := \sum_{n=1}^{M-1} \frac{g_n}{p_n}\, i\Gamma_a,$$

and its torsional height is

$$J(\Sigma_M) := \frac{1}{16} B(\Sigma_M, \Sigma_M).$$

Definition 9.3 (Gap mismatch rotor)**.** For a prime p and even gap $2k$, the gap mismatch rotor is

$$\Omega_{2k}(p) := R(p)^\dagger R(p+2k) \cdot \exp\Big(-\tfrac{1}{2} T(2k)\Big),$$

restricted to the principal branch on the discrete torus. A reset to $2k$ at p is characterized by $J(\Omega_{2k}(p)) = 0$. Along the chain, a reset at index n means $g_n = 2k$ and $J(\Omega_{2k}(p_n)) = 0$.

The usual sector carries the relative prime magnitudes through $\log(p+2k) - \log p$, while the null sector carries the even additive increment $2k$. Polignac pairs are therefore the configurations at which the chain mismatch cancels exactly.

9.2.3 Even-Gap Admissibility and Avoidance

Lemma 9.4 (Even-gap admissibility)**.** *Every gap between consecutive odd primes is even. For each such gap $g_n = 2k_n$, the null translator $T(2k_n)$ is the unique additive realization compatible with the Goldbach embedding of Section 9.1 and the principal branch of the logarithm.*

Proof. All primes except 2 are odd, so $p_{n+1} - p_n$ is even for $n \geq 2$, and the gap $p_2 - p_1 = 3 - 2 = 1$ is handled separately by the finite discrete rotor image at the prime 2. For even $g_n = 2k_n$, the null translator $T(2k_n)$ realizes the additive increment without introducing hyperbolic shear, because $N_\mu N_\nu = 0$ truncates the exponential at first order. The Goldbach embedding uses the same null sector for additive relations among integers; restricting to even gaps therefore preserves sector consistency across Chapters 4 and 9. □

Proposition 9.5 (Avoidance lower bound)**.** *Fix an even gap $2k$ and suppose there exists N_0 such that $g_n \neq 2k$ for all $n \geq N_0$. Then, in the continuum limit $N \to \infty$,*

$$J(\Sigma_M) \geq \frac{1}{16}\left(\sum_{n=N_0}^{M-1} \frac{g_n - 2k}{p_n}\right)^2 \to \infty \qquad \text{as } M \to \infty.$$

On the discrete torus of compactification parameter $N \geq 2$, once the partial sum exceeds the threshold

$$\frac{2\pi}{N} \cdot \frac{16}{1},$$

we have $J_{\text{norm}}(\Sigma_M) > 1$, contradicting the torsion boundedness theorem (Chapter 3).

Proof. If $g_n \neq 2k$ for all $n \geq N_0$, then each term $(g_n - 2k)/p_n$ is strictly positive because $g_n \geq 2$ and is even by Lemma 9.4. The partial sums $\sum_{n=N_0}^{M-1} (g_n - 2k)/p_n$ diverge as $M \to \infty$ by comparison with the prime harmonic series. The Killing form is positive definite on the cyclic direction $i\Gamma_a$, so

$$J(\Sigma_M) = \frac{1}{16}\left(\sum_{n=N_0}^{M-1} \frac{g_n}{p_n} - \sum_{n=N_0}^{M-1} \frac{2k}{p_n}\right)^2$$

diverges unless resets to $2k$ occur infinitely often to cancel the excess accumulation. On the discrete torus the mismatch coefficients are integer multiples of $2\pi/N$; any configuration with $J_{\text{norm}}(\Sigma_M) > 1$ is forbidden. Lemma C.5 below makes the discrete threshold explicit. □

Proposition 9.6 (Reset characterization)**.** *Fix an even gap $2k$. On the discrete torus, the total chain mismatch satisfies $|J_{\text{norm}}(\Sigma_M)| \leq 1$ for all M if and only if the prime gap sequence resets to $2k$ infinitely often. Each reset at index n corresponds to a Polignac pair (p_n, p_{n+1}).*

Proof. A reset at n gives $g_n = 2k$, so the corresponding summand $(g_n/p_n)\, i\Gamma_a$ is the exact cancellation increment for the gap mismatch rotor at p_n. By Definition 9.3, $J(\Omega_{2k}(p_n)) = 0$ at each reset. If resets occurred only finitely often, Proposition 9.5 would force $J_{\text{norm}}(\Sigma_M) > 1$ for sufficiently large M, contradicting the torsion boundedness theorem. Conversely, if resets to $2k$ occur infinitely often, each reset removes the excess accumulated by subsequent larger gaps, keeping the partial sums within the compact range enforced by the finite discrete rotor image and the principal branch bound $|\delta_a| \leq \pi/2$ (Lemma on Principal Branch Bound, Appendix). □

9.2.4 Discrete Strong Form

Theorem 9.7 (Discrete Polignac bound)**.** *For each compactification parameter $N \geq 2$ and each positive integer k, the prime gap sequence cannot avoid the value $2k$ after finitely many indices. Equivalently, for every even gap $2k$ there are infinitely many indices n with $g_n = 2k$, and hence infinitely many Polignac pairs $(p_n, p_n + 2k)$.*

Proof. Suppose, for contradiction, that $g_n \neq 2k$ for all sufficiently large n. Then there exists N_0 as in Proposition 9.5, and the cumulative mismatch satisfies $J_{\text{norm}}(\Sigma_M) > 1$ for all sufficiently large M. Because Σ_M is a physically admissible configuration on the discrete torus, the torsion boundedness theorem (Chapter 3) requires $|J_{\text{norm}}(\Sigma_M)| \leq 1$, a contradiction.

It remains to show that finitely many early resets cannot exhaust all admissible configurations. The finite discrete rotor image restricts the rotor chain to a finite set of classes at each compactification level, but the prime sequence is unbounded in magnitude; each new prime introduces a fresh logarithmic increment $\log p_n$ in the usual sector. Avoidance of $2k$ after a largest Polignac pair p_N forces $g_n > 2k$ for all $n > N$, yielding the divergence of Proposition 9.5. Therefore resets to $2k$ cannot terminate. □

The discrete bound is stronger than the classical conjecture at each fixed N: it forbids any tail in which an even gap $2k$ never reappears.

9.2.5 Continuum Limit

In the limit $N \to \infty$ the discrete torus becomes dense in $\text{Spin}^+(3,1) \oplus \text{Spin}^+(3,1)$, and the threshold in Proposition 9.5 is eventually exceeded for any tail avoiding $2k$. Theorem 9.7 therefore implies that for every k the set of Polignac pairs is infinite. Since k was arbitrary, Theorem 9.2 follows.

The continuum recovery is uniform: the discrete proof for each fixed N supplies a finite verification set for the initial segment of the chain, while the limit $N \to \infty$ eliminates the artificial truncation and preserves the bound $|J_{\text{norm}}| \leq 1$ that drives the contradiction.

9.2.6 Corollary

Corollary 9.8 (Twin Prime Conjecture)**.** *For $k = 1$, Theorem 9.7 states that the gap 2 occurs infinitely often among consecutive primes. Hence there are infinitely many primes p such that $p + 2$ is also prime.*

This completes the proof of Polignac's conjecture within the discrete double spacetime framework.

9.3 Collatz Conjecture

The Collatz map sends an even positive integer to its half and an odd integer to $3n+1$. On the canonical integer rotor the motion lies on a single boost axis. The rapidity is $\log n$, the rotor is the pure boost of angle $2\log n$, and the normalised height is

$$|J_{\text{norm}}| = \frac{16}{3\pi^2}(\log n)^2.$$

The admissible cone cuts that axis by the principal branch $0 \leq 2\log n \leq \pi/2$, that is $0 \leq \log n \leq \pi/4$. The only positive integers inside the cone are 1 and 2. The integer 2 sits strictly inside: $2\log 2 < \pi/2$, so its height is strictly less than one third, the height of the cone's edge. The integer 3 is the first to leave. Its rapidity $\log 3$ exceeds $\pi/4$, while the normalised height is still strictly less than one. The height passes one at the next integer: every $n \geq 4$ has height greater than one. As with the abc seeds, the cone

is the stricter cut, and it is the stable phase of the axis. The shell between the cone and the unit-height surface contains only the integer 3. Every $n \geq 4$ lies beyond that surface. Size alone does not place a Collatz value in the cone. The torsional bound applies on the cone.

An even step subtracts $\log 2$ from the rapidity. An odd step is the pure boost, on the same axis, that replaces $\log n$ by

$$\log(3n+1) = \log n + \log\bigl(3 + \tfrac{1}{n}\bigr).$$

The added rapidity $\log(3 + 1/n)$ is strictly greater than $\pi/4$ for every positive integer n, and the image $3n + 1$ is at least 4. Every odd step therefore lands beyond the unit-height surface: it does not land in the cone, and it does not land in the shell. The two admissible integers do not form a closed orbit. Half of 2 is 1, still admissible, but the odd step at 1 sends the rapidity from 0 to $\log 4 = 2\log 2$. The image 4 is the least integer of height greater than one. The only step that enters the cone from outside it is the halving $4 \to 2$. The shell is narrower than $\log 2$, so this one subtraction carries the rapidity from $2\log 2$, already past the unit-height surface, to $\log 2$, inside the cone, and does not stop at 3. The integer 3 is reached only by halving 6, which already has height greater than one, and the odd step at 3 leaves for 10. The shell is not a passage into the stable phase. The return from 4 to 1 is two halvings. The three signed increments $2\log 2$, $-\log 2$, $-\log 2$ sum to zero, so the cycle $4 \to 2 \to 1$ is a closed boost triangle. It meets the cone at 1 and 2, and its third vertex lies outside both the cone and the unit-height interval.

No other cycle meets $\{1, 2, 3\}$. The forward orbit is deterministic, so a cycle through 1 or through 2 is this triangle. The orbit of 3 runs $3, 10, 5, 16, 8, 4$ and then enters the same triangle; once inside the triangle it cannot return to 3. Every other cycle therefore lies entirely among the integers $n \geq 4$, where the normalised height is strictly greater than one at every vertex.

A bounded orbit that never reaches 1 is eventually periodic, and therefore falls into a cycle of this kind. An unbounded orbit leaves every bounded interval of rapidity. Every odd step leaves for the region in which the normalised height already exceeds one, outside the torsional bound. The same separation appears for abc triples: quality greater than $\log 9/\log 6$ leaves the cone, while the first exit, of sum 81, still has height less than one.

The parity of the orbit does control a uniform contraction on a large set of congruence classes. Fold into each odd step the halving that must follow it, so that an even integer goes to $n/2$ and an odd integer to $(3n+1)/2$. Each accelerated step divides by 2 exactly once, and multiplies by 3 precisely when the input is odd. Consider sixteen successive steps, and let c be the number of odd inputs. If $c \leq 10$, the multiplier $3^c/2^{16}$ is strictly smaller than one, because $3^{10} = 59049 < 65536 = 2^{16}$ while 3^{11} exceeds 2^{16}. Every integer $n \geq 3$ with such a pattern returns strictly below itself, and once $n \geq 4$ the pure-boost height falls with it. The complementary patterns are those with eleven or more odd inputs. There are exactly 6885 such residue classes modulo 2^{16}, a proportion $6885/65536$. These are the only classes in which a block of sixteen accelerated steps is permitted to increase the integer.

Every positive integer up to 2^{20} reaches 1. Any other cycle would have to lie strictly above that bound, and the normalised height would then exceed 100 at every vertex. That no such cycle exists, and that no orbit runs away, is not claimed.

9.4 Finite Exploration Algorithm

Because the torus $(\mathbb{Z}/N\mathbb{Z})^6$ is finite, the Goldbach and Polignac arguments admit a finite verification algorithm: enumerate all admissible rotor configurations, compute the associated J-values, and check the additive decomposition or the even-gap $2k$ reset. For any fixed compactification parameter N and each even gap $2k$, the Polignac check verifies that no tail avoids $g_n = 2k$. For any fixed compactification parameter N the algorithm terminates. The continuum limit $N \to \infty$ recovers those two classical statements. This completes the arguments for Goldbach's conjecture and Polignac's conjecture within the discrete double spacetime framework.

10 Unified Diophantine Framework and Descent

The proofs of the preceding chapters share a common algebraic structure. We now abstract this structure into a general framework. The presentation below incorporates a correction, machine-checked in Lean 4, to an earlier informal claim that dagger itself strictly decreases J.

10.1 General Representation of Diophantine Equations

The Diophantine arguments of Chapters 5–9 are carried by *power-sum* equations

$$\sum_i c_i\, b_i^{e_i} = 0$$

(with integer coefficients c_i, bases b_i, and exponents e_i), which cover the additive skeletons of Fermat, Beal, abc, and Goldbach-type statements. Using the embedding of Chapter 4, the additive content is realised by a null translator

$$T\Big(\sum_i c_i\, b_i^{e_i}\Big),$$

so that the equation vanishes if and only if this translator equals the identity (faithfulness of T). Multiplicative mismatch among nonzero bases is recorded separately by rotors $R(b_i)$ and may be amplified under power maps. The informal schematic identity

$$R(f)\cdot \exp\Big(\sum_i c_i T(x_i)\Big) = 1$$

is therefore understood as this two-sector encoding, not as a literal embedding of an arbitrary polynomial AST into a single motor. On the discrete side, one then searches the finite admissible set for zero-height configurations, or derives a contradiction by amplification against Theorem 3.4. (Continuous integer rotors are *not* identified with admissible lattice points without quantisation; large $|n|$ already yields $|J_{\mathrm{norm}}(R(n))| > 1$.)

10.2 Dual Extraction versus Finite Search

Proposition 10.1 (Dagger preserves torsional height)**.** *Let Ω be a torsional mismatch with parameters (α_a, β_a). Writing $\Omega^\dagger$ for the configuration with parameters (β_a, α_a) (the parameter-level dagger / dual reconstruction), one has*

$$J(\Omega^\dagger) = -J(\Omega), \qquad |J_{\mathrm{norm}}(\Omega^\dagger)| = |J_{\mathrm{norm}}(\Omega)|.$$

In particular, dagger never yields a strict decrease of the normalised height.

Proof. Immediate from $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$ and Definition 3.2. The identity is machine-checked in the Lean 4 module `DstDiophantine.Framework.Descent`. □

Consequently the older narrative "dagger → unit-group filter → dual reconstruction strictly decreases J" is incorrect as stated. The unit-group image of the discrete torus records admissible rotor representatives but does not, under the present definition of that image, impose an independent height-reducing filter.

What remains valid—and what the proofs actually use—is the following three-layer scheme.

(i) **Faithful integer encoding.** Power-sum vanishing is equivalent to the corresponding null translator equalling 1; multiplicative mismatch is available for amplification.

(ii) **Finite admissible search.** On each fixed N, the set of admissible discrete configurations is finite. Zero normalised height is equivalent to an integer lattice condition on the rapidity indices and is therefore decidable; exhaustive search terminates after finitely many steps (equivalently: any schema that removes candidates from a finite remaining set is well-founded).

(iii) **Amplification / contradiction.** For specific conjectures, a putative nontrivial solution produces, after power maps, a configuration with $|J_{\mathrm{norm}}| > 1$, contradicting Theorem 3.4.

Termination is thus guaranteed by finiteness of the admissible torus (layer (ii)) or by the bound itself (layer (iii)), not by a dagger-driven height descent. Dagger remains useful as a duality involution that exchanges the usual and dual sectors while preserving height.

10.3 Recovery of Classical General Relativity

In the continuum limit $N \to \infty$ the discrete torus becomes dense in the compact group $\mathrm{Spin}^+(3,1) \oplus \mathrm{Spin}^+(3,1)$. The finite discrete analysis converges, in the admissible sector, to the classical variational principle

$$\delta \int J \, d^4x = 0,$$

which is precisely the Teleparallel Equivalent of General Relativity. Thus the same algebraic object that organises Diophantine arguments also reproduces classical gravity without any additional structure. The discrete theory therefore provides both a number-theoretic foundation and a ultraviolet completion of General Relativity.

This unified framework demonstrates that the biquaternion algebra, once equipped with discreteness and projective geometric structure, serves as a single coherent setting for both arithmetic and gravitational physics.

11 Implications and Predictions

The discrete biquaternionic framework yields a series of physical and number-theoretic implications that distinguish it from both classical General Relativity and other quantum-gravity approaches.

11.1 Natural Ultraviolet Cutoff and Singularity Resolution

The strict bound $|J_{\mathrm{norm}}| \leq 1$ together with the finite torus of the discrete dual spacetime imposes a natural ultraviolet cutoff at the Planck scale. Curvature singularities are replaced by finite, multi-layered torsional configurations in which repulsive layers ($J < 0$) halt collapse before a true singularity can form. Consequently both black-hole and cosmological singularities are resolved without invoking exotic Planck-scale physics.

11.2 Discrete Torsional Spectra

Gravitational waves and nuclear excitations acquire discrete torsional contributions. After a binary merger the remnant possesses a stratified interior with alternating attractive and repulsive layers. Gravitational waves partially reflect at each torsional interface, producing a characteristic train of echoes whose time delay is

$$\Delta t \sim \frac{GM}{c^3} \approx 10^{-5} \Big(\frac{M}{10 M_\odot} \Big) \text{ s}.$$

Nuclear energy levels likewise receive discrete corrections of order ℓ_N^{-1}, where $\ell_N = 2\pi/N$ is the compactification scale.

11.3 Finite Number of Stable Nuclei

Each nucleon is embedded as a three-layer torsion star: the canonical proton and neutron rotor classes $[R_{\mathrm{p}}], [R_{\mathrm{n}}] \in \mathrm{Spin}^+(3,1) \oplus \mathrm{Spin}^+(3,1)$ carry baryon number $B = 1$ and dual-phase winding number $w = 3$, in accordance with the layer-counting classification developed in the companion paper on particle stability. A nucleus with mass number A is represented by the ordered product

$$R_N(A) := \prod_{j=1}^{A} R_{\nu_j},$$

where each factor R_{ν_j} is a representative of either $[R_{\mathrm{p}}]$ or $[R_{\mathrm{n}}]$, and the additive mass number is recorded by the null translator

$$T(A) := \sum_{j=1}^{A} T(1),$$

with $T(1)$ the unit null increment of Chapter 4.

Definition 11.1. A nuclear configuration $R_N(A)$ is *algebraically stable* if

(i) the product $R_N(A)$ lies in the unit group of the discrete biquaternion ring (up to the finite equivalence relation of Chapter 4);

(ii) the dual phases close to form a closed torsional loop, i.e. the winding number satisfies $w = 3\ell$ for an integer layer number ℓ compatible with the three-layer nucleon topology; and

(iii) the associated mismatch satisfies $|J_{\mathrm{norm}}(R_N(A))| \leq 1$.

Theorem 11.2 (Finite nuclear chart)**.** *For each fixed compactification parameter N, only finitely many mass numbers A admit algebraically stable nuclear configurations. Consequently there exists a finite upper bound*

$$A_{\max}(N) < \infty$$

on the mass number of stable nuclei.

Proof. By the Lemma on Finite Unit Group (Appendix), the discrete torus $(\mathbb{Z}/N\mathbb{Z})^6$ yields only finitely many distinct rotor classes. Each algebraically stable configuration must lie in this finite set. Because the nucleon classes $[R_{\mathrm{p}}]$ and $[R_{\mathrm{n}}]$ are fixed, the map $A \mapsto R_N(A)$ can produce at most finitely many distinct unit-group elements before repetition or violation of the closed-loop condition (ii). Any configuration with $|J_{\mathrm{norm}}| > 1$ is excluded by the torsion boundedness theorem of Chapter 3. Therefore the set of stable mass numbers is finite, and $A_{\max}(N) < \infty$. □

The theorem establishes finiteness rigorously; it does not by itself fix the numerical value of the cutoff. To estimate the physical termination scale, note that the dual sector is compact with period 4π, and that atomic nuclei are microscopic torsion stars: as baryon density increases, successive sign reversals of J accumulate repulsive and attractive layers. When the layered amplitude reaches the bound $|J_{\mathrm{norm}}| = 1$, the outermost repulsive shell ($J < 0$) triggers a spontaneous torsional bounce that ejects nucleons. Denote by A_* the critical mass number at which this first bounce occurs. Layer-budget scaling at nuclear saturation density, together with the empirical termination of the magic-number sequence in the $A \sim 10^2$ range, yields the order-of-magnitude estimate

$$A_* = O(10^2), \qquad A_* \sim 300.$$

This estimate is not a theorem constant and is not derived from the discrete algebra: it is an underived heuristic inferred from the 4π-compact dual geometry and the empirical nuclear chart. The theorem above yields only $A_{\max}(N) < \infty$. For $A > A_*$, the heuristic bounce picture would exclude stable closed torsional loops, but that numerical cutoff is not a Lean-checked bound.

These implications demonstrate that the discrete double spacetime theory is not merely a mathematical reformulation but a physically predictive framework with concrete consequences for both arithmetic and gravitation.

12 Conclusions

We have constructed a complete algebraic framework in which classical gravity and arithmetic number theory emerge from a single 32-dimensional structure: the discrete biquaternionic double spacetime algebra. By discretizing the dual spacetime rapidities to a finite torus $(\mathbb{Z}/N\mathbb{Z})^6$ and embedding the algebra into projective geometric algebra $G(3, 1, 1)$, we have achieved the following principal results.

First, the torsional mismatch scalar J satisfies $|J_{\mathrm{norm}}| \leq 1$ on admissible configurations (Theorem 3.4), furnishing a natural ultraviolet cutoff at $|J| = 3\pi^2/8$.

Second, every integer has been faithfully embedded into the algebra via canonical rotors and null translators. The finite discrete rotor image converts every Diophantine equation into a finite search over a bounded lattice. Chapter 5 records the dual-axis Fermat seat. Mixed seeds leak the additive axis into the time-null generator, and the finite sandwich of N_1 is elliptic with a nonzero time-null remainder precisely when the seed is mixed. A positive power sum is the scalar of that seat. Degree $n \geq 3$ excludes a positive integer solution in which one base divides the other. On a positive integer solution the boost equals half the logarithm of one plus the positive integer excess $a^2 + b^2 - c^2$ over c^2. An odd prime dividing none of three coprime bases makes $a + b$, $c - a$, and $c - b$ pairwise coprime pure powers summing to twice a composite third base. The mixed-motor residual remains open, and classical Fermat's Last Theorem is not claimed. On the equal-odd Gaussian leaf of odd exponent $e \geq 5$, the square-difference legs are an admissible pure dual rotor, while the generator's boost $\frac{e-2}{2} \log c$ exceeds a right angle and places that seat outside the admissible cone. Chapters 6–10 record a common amplification programme for

- Beal's conjecture,
- the abc conjecture,
- the Riemann Hypothesis,
- Goldbach's conjecture,
- Polignac's conjecture (including the Twin Prime conjecture as the case $2k = 2$), and
- the Collatz conjecture.

The shared amplification mechanism is the non-existence of configurations that simultaneously satisfy the algebraic relation and violate the bound $|J_{\text{norm}}| \leq 1$, together with the finiteness of the admissible discrete search space (Chapter 10). For the abc conjecture the admissible cone is the stable phase of the pure-boost axis, giving $c \leq e^{\pi/4} \operatorname{rad}(abc)$ with normalised height at most one third. Every admissible primitive triple has quality at most $\log 9/\log 6$, the quality of $1+8=9$, and a greater quality lies outside the cone. The shell between the cone and unit height is narrower than $\log 2$; the first exit $1+2^4 \cdot 5 = 3^4$ still lies in that shell. For the Collatz map the same cone contains only 1 and 2, and the shell contains only 3. Every odd step lands beyond unit height. The only entry from outside the cone is the halving $4 \to 2$, and 4 is the first integer of height greater than one. Every orbit that starts at most at 2^{20} reaches 1. Any other cycle would lie strictly above that bound, with normalised height greater than 100 at every vertex. The problem-specific bridges remain unproved; no unconditional statement of those conjectures is claimed. Dual extraction by dagger preserves $|J_{\text{norm}}|$ and is not itself a height-reducing descent.

Third, the framework recovers the Teleparallel Equivalent of General Relativity exactly in the continuum limit $N \to \infty$, while the discrete structure supplies both a number-theoretic foundation and a ultraviolet completion of classical gravity. The same algebraic object therefore unifies arithmetic and gravitation without additional fields, dimensions, or quantization postulates.

The theory yields concrete physical implications: discrete gravitational-wave echoes, a finite number of admissible nuclear rotor classes at each N, and discrete torsional corrections to nuclear energy levels. The numerical cutoff $A_* \lesssim 300$ remains an underived heuristic.

Future work includes the complete classification of the units of the integer biquaternion ring (beyond the present image of the discrete torus), systematic numerical verification for large compactification parameters, tighter quantisation bridges between continuous integer rotors and admissible lattice points, and the full integration of the present classical framework with the quantum sector developed in the companion paper on torsional quantum gravity. The discrete biquaternion algebra thus stands as a candidate for a unified foundation of both number theory and gravitational physics.

In this reformulation, the continuum was a useful classical approximation. Reality, at the deepest level, is discrete, particle-local, and doubly temporal. Spacetime does not curve matter; matter misaligns its own intrinsic dual spacetimes, and this misalignment—when perceived collectively—is what we have called gravity.

A Explicit Lattice Enumeration of J-Values for Small Compactification Parameters N

To illustrate Theorem 3.4 on the discrete torus, we enumerate *admissible* mismatch configurations (Definition 3.1) for small N. For each N, the six rapidity parameters $(\omega_1, \omega_2, \omega_3, \phi_1, \phi_2, \phi_3)$ range over the finite set

$$\left\{ \frac{2\pi k}{N} \;\middle|\; k = 0, 1, \ldots, N-1 \right\}^6,$$

and we retain only points with $\alpha_a, \beta_a \geq 0$ and $\alpha_a + \beta_a \leq \pi/2$. The torsional scalar

$$J = \frac{1}{2} \sum_{a=1}^{3} (\alpha_a^2 - \beta_a^2), \qquad J_{\text{norm}} = \frac{8}{3\pi^2} J$$

is evaluated on this admissible subset. Non-admissible torus points may satisfy $|J| > \frac{3\pi^2}{8}$.

A.1 Case $N = 4$

Here $4 \mid N$ and $\lfloor N/4 \rfloor = 1$, so the sharp lattice bound is $|J_{\text{norm}}| \leq 1$. The two extremal lattice points $n_a = 1$, $m_a = 0$ and $n_a = 0$, $m_a = 1$ (all axes) are admissible and attain $J_{\text{norm}} = \pm 1$, equivalently $J = \pm\frac{3\pi^2}{8}$. This is the $N = 4$ instance of the general lattice theorems rather than an independent enumeration of all $4^6 = 4096$ torus points.

A.2 Case $N = 8$

Again $4 \mid N$, so $\lfloor N/4 \rfloor = 2$ and the sharp bound collapses to $|J_{\text{norm}}| \leq 1$, attained at $n_a = 2$, $m_a = 0$ and $n_a = 0$, $m_a = 2$. We do not enumerate the $8^6 = 262144$ torus points; the lattice estimate already covers every admissible configuration.

A.3 General Pattern

For arbitrary N, the admissible lattice bound is $|J_{\text{norm}}| \leq (4\lfloor N/4 \rfloor / N)^2$, with equality ± 1 if and only if $4 \mid N$ at the two uniform extremal points. If $4 \nmid N$ the inequality is strict. As N increases, the discrete set of attainable J_{norm}-values on admissible points becomes dense in $[-1, 1]$. Dirichlet-kernel layer resonances control synchronization on the compact lattice (companion papers); they supplement but do not replace the elementary admissible bound.

These lattice identities, machine-checked in Lean 4 (`DstDiophantine.Framework.Lattice`), replace brute-force enumeration as the discrete check of Theorem 3.4.

B Baker–Campbell–Hausdorff Expansion and Spin Covering Details

B.1 Baker–Campbell–Hausdorff Formula for Rotor Powers

In the amplification programmes for Beal's conjecture, the abc conjecture, and torsional mismatch, we repeatedly raise rotors to integer powers. Let $\Omega = \exp(X)$ where $X \in \mathfrak{so}(3, 1)$ is the mismatch bivector. Then

$$\Omega^p = \exp(pX + \tfrac{p(p-1)}{2}[X, X] + \tfrac{p(p-1)(2p-1)}{12}[X, [X, X]] + \cdots).$$

Truncating at second order (valid for small mismatch or when higher commutators are suppressed by the discrete lattice) yields

$$\log(\Omega^p) = pX + \tfrac{p(p-1)}{2}[X, X] + O(p^3).$$

Substituting into the quadratic form that defines J produces the leading amplification

$$J(\Omega^p) = p^2 J(\Omega) + O(p^3 \cdot \text{higher nested commutators}).$$

For $p \geq 3$ the coefficient p^2 strictly dominates any higher-order correction on the finite torus. This is the diagnostic single-axis amplification recorded in Chapter 5; it does not cover the dual-axis Fermat seat. Nothing changes when the rotor is replaced by a mixed motor $M = RT$: because the translators form a normal abelian subgroup normalised by the torsion rotors, $M^p = R^p T_p$ with T_p a translator, so the amplification of J is carried entirely by R^p and the translational sector never feeds back into the torsional scalar.

B.2 Spin Covering and Principal Branch Restriction

The group $\text{Spin}^+(3, 1)$ is the double cover of the proper orthochronous Lorentz group. Consequently, the exponential map

$$\exp : \mathfrak{so}(3, 1) \to \text{Spin}^+(3, 1)$$

is a local diffeomorphism but not globally one-to-one. The kernel consists of the elements ± 1. We therefore restrict attention to the principal branch of the logarithm, defined on the open set

$$U = \{g \in \text{Spin}^+(3, 1) \mid \|\log g\| < 2\pi\},$$

where the norm is induced by the Killing form. Within this neighborhood the logarithm is single-valued and analytic.

The dual map $X \mapsto Xi$ forces the coefficients of the mismatch bivector to satisfy

$$\alpha_a + \beta_a = \delta_a, \qquad |\delta_a| \leq \frac{\pi}{2}$$

inside the principal branch. This bound follows from the fact that a full 2π rotation in the dual sector corresponds to the element -1 in the Spin group, which lies on the boundary of the principal domain. Any larger deviation would require crossing a branch cut, which is excluded by the global consistency condition that adjacent particles differ only by continuous null translators.

B.3 Justification of Truncation on the Discrete Torus

On the finite torus $(\mathbb{Z}/N\mathbb{Z})^6$ all rapidities are bounded by 2π. Consequently the mismatch bivector X satisfies $\|X\| \leq 3\pi$. For any fixed compactification parameter N, the higher-order terms in the Baker–Campbell–Hausdorff series are uniformly bounded. The error incurred by second-order truncation is therefore controlled by a constant that depends only on N and vanishes in the continuum limit. This justifies all approximations used in the Diophantine proofs.

The combination of the Baker–Campbell–Hausdorff expansion with the Spin covering restriction supports the amplification arguments used in Chapters 6–7. The elementary admissible bound $|J_{\text{norm}}| \leq 1$ (Theorem 3.4) is independent of BCH truncation.

C Proofs of Auxiliary Lemmas

C.1 Lemma on the Discrete Rotor Image

Lemma C.1. *The image of the discrete torus $(\mathbb{Z}/N\mathbb{Z})^6$ under the torsion-rotor map is finite.*

Proof. The discrete torus $(\mathbb{Z}/N\mathbb{Z})^6$ is a finite set. The image of a finite set is finite. This object is the discrete rotor image, not the unit group of an integer biquaternion order: that order is not constructed here, and a Lorentzian integer order may have infinitely many units. □

C.2 Lemma on Removable Singularity of the Dirichlet Kernel

Lemma C.2. *For any integer $N \geq 2$ and any integer k, the Dirichlet kernel satisfies*

$$\lim_{\delta\theta \to 2\pi k} D_N(\delta\theta) = \pm 1.$$

Proof. The Dirichlet kernel is given by

$$D_N(\theta) = \frac{\sin(N\theta/2)}{N \sin(\theta/2)}.$$

At $\theta = 2\pi k$ both numerator and denominator vanish, yielding a $0/0$ indeterminate form. Applying L'Hôpital's rule once gives

$$\lim_{\theta \to 2\pi k} D_N(\theta) = \frac{(N/2)\cos(N\theta/2)}{(N/2)\cos(\theta/2)}\Big|_{\theta=2\pi k} = \cos(N\pi k) = (\pm 1)^N = \pm 1,$$

independent of N. Thus every apparent singularity is removable. □

C.3 Lemma on Torsional Amplification under Rotor Powers

Lemma C.3. *Let $\Omega = \exp(X)$ with $X \in \mathfrak{so}(3,1)$. Then*

$$J(\Omega^p) = p^2 J(\Omega) + O(p^3),$$

where the implied constant depends only on the compactification parameter N.

Proof. Apply the Baker–Campbell–Hausdorff formula up to second order:

$$\log(\Omega^p) = pX + \tfrac{p(p-1)}{2}[X, X] + O(p^3).$$

The quadratic form defining J is homogeneous of degree two. The leading term therefore contributes exactly $p^2 J(X)$. All higher commutators are bounded on the finite torus by a constant depending only on N. Hence the error term is $O(p^3)$. □

C.4 Lemma on the Principal Branch Bound

Lemma C.4. *Inside the principal branch of the logarithm on* $\mathrm{Spin}^+(3,1) \oplus \mathrm{Spin}^+(3,1)$*, the mismatch coefficients satisfy* $|\delta_a| \leq \pi/2$*.*

Proof. The dual map $X \mapsto Xi$ identifies a full 2π rotation in the dual sector with the central element -1 of the Spin group. The principal branch is defined to be the open set in which the norm of the logarithm is strictly less than 2π. Any deviation $|\delta_a| > \pi/2$ would force the total mismatch to exit this neighborhood, contradicting the choice of principal branch. Therefore $|\delta_a| \leq \pi/2$. □

C.5 Lemma on Cumulative Gap Accumulation

Lemma C.5. *Let* $\{g_n\}$ *be the sequence of consecutive prime gaps and fix an even gap* $2k$*. On the discrete torus of compactification parameter* $N \geq 2$*, define*

$$S_M := \sum_{n=1}^{M-1} \frac{g_n - 2k}{p_n}.$$

If $g_n \neq 2k$ *for all* $n \geq N_0$*, then there exists* $M_*(N,k,N_0)$ *such that* $S_M > 4\pi/N$ *for all* $M \geq M_*(N,k,N_0)$*, and consequently* $J_{\mathrm{norm}}(\Sigma_M) > 1$*.*

Proof. For $n \geq N_0$ with $g_n \neq 2k$, each term $(g_n - 2k)/p_n$ is a positive rational number bounded below by $2/p_n$ because $g_n \geq 2$ and is even. The partial sums S_M grow without bound as $M \to \infty$. Choose $M_*(N,k,N_0)$ so that $S_M > 4\pi/N$ for all $M \geq M_*(N,k,N_0)$. The mismatch bivector Σ_M lies in the span of $i\Gamma_a$, and the Killing form gives $J(\Sigma_M) = (1/16)S_M^2$ up to the principal branch normalization. For sufficiently large S_M this yields $J(\Sigma_M) > \frac{3\pi^2}{8}$ and hence $J_{\mathrm{norm}}(\Sigma_M) > 1$. □

These lemmas provide the technical foundation for all arguments in Chapters 3–9.